

Problem 4, Cell 6(a): the sizes $k = 32, 38, 42$ (computer-assisted finite check) ($k = 44$: evidence only, not claimed)

Official statement (Cell 6). “(a) Decide a size $k \geq 25$.”

Throughout, “decide” is used as in Cell 5: to decide a size k means to show that k is not the least size at which the conjecture fails.

Conjecture (Sun). If $k \geq 2$ residue classes $a_1 \bmod m_1, \dots, a_k \bmod m_k$ are pairwise disjoint, then $\gcd(m_i, m_j) \geq k$ for some $i \neq j$. We say the conjecture *holds at k* if this is true for that k .

Result.

Theorem 1. *Let $k \in \{32, 38, 42\}$. If the conjecture holds at $k - 1$, then it holds at k .*

Remark 2 ($k = 44$: computer-assisted evidence, **not claimed**). Section 4 reduces the step “the conjecture at 43 implies it at 44” to 481 finite configurations, and a SAT solver answers UNSAT on all of them (`sat_residual.log`). An exhaustiveness certificate in the sense of the hand-in rules (in particular the smallest forcing part for each configuration) has *not* been completed, so we do **not** claim $k = 44$.

Corollary 3. *If the conjecture fails for some k , the least such k is not 32, 38 or 42.*

Proof. If k is the least failing size, the conjecture holds at $k - 1$, and Theorem 1 gives a contradiction. □

Status of the hypothesis. The corollary is unconditional. Theorem 1 by itself does *not* establish the conjecture at 32, 38 or 42, because the hypothesis at $k - 1 = 31, 37, 41$ is **not** established by our certified computation. That computation covers $k - 1 \leq 16$ only. O’Bryant (2007, Theorem 3) states the conjecture for $k \leq 20$; we neither assume nor use that result, and it also stops below 31.

$k = 48$. Not decided; see Remark 4.

Credit. The method is O’Bryant’s: *On Z.-W. Sun’s disjoint congruence classes conjecture* (2007, arXiv:math/0604347), proof of Lemma 6. Cell 5 used it for $k = 24, 30$. That argument needs three distinct primes dividing one gcd to force $\gcd \geq 30 > p$. This fails once $p = k - 1 \geq 31$, since $\gcd = 30$ is then allowed. The new ingredients are:

- two-coordinate projections in place of the full injection (Lemma 8);
- a residue-counting bound on the classes whose gcd with each other is exactly 30 (Lemma 10);

- an exact enumeration of the resulting integer inequalities over all distributions of the primes (script `check_p4_c6.py`);
- for $k = 44$, a sharper joint count for the two light triples $\{2, 3, 5\}$ and $\{2, 3, 7\}$ (Lemma 13), and a SAT check of the residual configurations (`sat_residual.py`).

1 Tools

Lemma 4. $a \bmod m$ and $b \bmod n$ are disjoint if and only if $\gcd(m, n) \nmid a - b$.

Proof. If x lies in both, then $a \equiv x \equiv b \pmod{\gcd(m, n)}$. Conversely, suppose $g = \gcd(m, n)$ divides $a - b = gt$, and take $um + vn = g$. Then $x = a - umt = b + vnt$ lies in both. $\square$

Fix a pairwise disjoint family $a_i \bmod m_i$ ($i \in [k]$) and write $g_{ij} = \gcd(m_i, m_j)$. Let $\text{pr}(n)$ be the set of primes dividing n . By Lemma 4, every $g_{ij} \geq 2$ and $a_i \not\equiv a_j \pmod{g_{ij}}$. Put $\pi(i) = \{q \text{ prime} : q \mid g_{ij} \text{ for some } j \neq i\}$. Every prime in $\pi(i)$ divides m_i .

Lemma 5. For $i \neq j$, $\text{pr}(g_{ij}) = \pi(i) \cap \pi(j)$. In particular $\pi(i) \cap \pi(j) \neq \emptyset$, and $\prod_{q \in \pi(i) \cap \pi(j)} q$ divides g_{ij} .

Proof. $\subseteq$ is clear. If $q \in \pi(i) \cap \pi(j)$, then $q \mid m_i$ and $q \mid m_j$, so $q \mid g_{ij}$. The last claim holds because distinct primes dividing g_{ij} have their product dividing g_{ij} . $\square$

2 Setting

Throughout, $k \in \{32, 38, 42\}$ and $p = k - 1 \in \{31, 37, 41\}$ is prime. Let r be the number of primes below p , so $r = 10, 11, 12$. Suppose, for contradiction, that the conjecture holds at $k - 1$, that the family above is pairwise disjoint, and that $g_{ij} \leq p$ for all $i \neq j$.

The set S (as in Cell 5, Steps 1–4). Let $S = \{i : p \in \pi(i)\}$ and $s = |S|$.

1. For $i \neq j$ in S we have $p \mid g_{ij} \leq p$, so $g_{ij} = p$ and $\pi(i) \cap \pi(j) = \{p\}$ (Lemma 5). A pair has gcd p only if both members lie in S .
2. $s \geq 3$. Remove any index x . The remaining $k - 1$ classes are pairwise disjoint, so by the hypothesis at $k - 1$ some pair among them has $\gcd \geq k - 1 = p$, hence exactly p . That pair lies in $S \setminus \{x\}$. So $|S \setminus \{x\}| \geq 2$ for every x , and $s \geq 3$.
3. $S \neq [k]$. Otherwise the k residues a_i would be pairwise incongruent modulo p , which is impossible since $k > p$. Put $O = [k] \setminus S$, so $|O| = k - s \geq 1$.
4. Number $S = \{i_1, \dots, i_s\}$ and let $P_j = \pi(i_j) \setminus \{p\}$. By item 1 the P_j are pairwise disjoint. They consist of primes $< p$, because each prime in $\pi(i_j)$ divides some $g \leq p$. For $l \in O$, $p \notin \pi(l)$, since otherwise $l \in S$. So by Lemma 5,

$$\text{pr}(g_{l, i_j}) = \pi(l) \cap \pi(i_j) = \pi(l) \cap P_j \neq \emptyset. \quad (1)$$

In particular every P_j is nonempty (as $O \neq \emptyset$), $\sum_j |P_j| \leq r$, and $s \leq r$.

Call $P_1, \dots, P_s$ the *blocks*. Since the blocks are disjoint and nonempty and lie in the set of r primes below p , any s' of them have total size at most $r - (s - s')$.

Choice functions. A *choice function* is a map $\omega : O \rightarrow P_1 \times \cdots \times P_s$ with $\omega_j(l) \in \pi(l) \cap P_j$ for all l, j . Such a map exists by (1).

Lemma 6 (agreement). *Let ω be a choice function, let $l \neq l'$ be in O , and let $A = \{j : \omega_j(l) = \omega_j(l')\}$. Then $\prod_{j \in A} \omega_j(l)$ divides $g_{ll'}$. In particular it is $\leq p$.*

Proof. For $j \in A$ the prime $q = \omega_j(l)$ lies in $\pi(l) \cap \pi(l')$, so $q \mid g_{ll'}$ (Lemma 5). These primes are distinct because the blocks are disjoint. $\square$

Call a set of distinct primes *light* if its product is $\leq p$.

Lemma 7 (light sets, $p \in \{31, 37, 41\}$). *1. No light set has 4 or more elements. The only light 3-set is $\{2, 3, 5\}$.*

2. A light 2-set of primes ≥ 5 is $\{5, 7\}$, and this set is light only when $p \geq 35$.

3. Every light set containing a prime q with $2q > p$ is $\{q\}$.

Proof. $2 \cdot 3 \cdot 5 \cdot 7 = 210 > p$ and $2 \cdot 3 \cdot 7 = 42 > p$. Every other 3-set has product $\geq 2 \cdot 3 \cdot 11 > p$ or $\geq 2 \cdot 5 \cdot 7 > p$. For (2): $5 \cdot 11 = 55 > p$, so only $5 \cdot 7 = 35$ can be light. For (3): the product is at least $2q > p$. (The script re-checks (1) and (2) mechanically.) $\square$

Lemma 8 (projection). *Let ω be a choice function and let $j \neq j'$. Suppose $l \neq l'$ in O satisfy $\omega_j(l) = \omega_j(l') = x$ and $\omega_{j'}(l) = \omega_{j'}(l') = y$. Then $\{x, y\}$ is light. If, in addition, they agree in a third coordinate z , then $\{x, y, z\} = \{2, 3, 5\}$ and $g_{ll'} = 30$.*

Proof. Apply Lemma 6, then Lemma 7. In the last case $30 \mid g_{ll'} \leq p < 60$. $\square$

3 Proof of Theorem 1

We show $|O| < k - s$ in every case, which contradicts $|O| = k - s$.

Case I: two blocks avoiding 2 and 3 (all $s \geq 4$, and $s = 3$ except Case II)

Lemma 9. *Suppose two blocks P_α, P_β contain neither 2 nor 3. Then:*

- 1. if $\{5, 7\}$ is not split between P_α and P_β , or $p < 35$, then $|O| \leq |P_\alpha||P_\beta|$;*
- 2. otherwise $|O| \leq |P_\alpha||P_\beta| - 1 + |P_\gamma|$ for every third block P_γ .*

Proof. Take any choice function and project $l \mapsto (\omega_\alpha(l), \omega_\beta(l))$. Both coordinates are primes ≥ 5 . By Lemmas 8 and 7(2), two distinct elements of O with the same image (x, y) need $\{x, y\} = \{5, 7\}$ and $p \geq 35$. This proves (1). In case (2), only the fibre over the pair $(5, 7)$ can have more than one element. Two elements of that fibre cannot also agree in coordinate γ , by Lemma 8, because $\{5, 7, z\}$ is never light. So the fibre injects into P_γ , and $|O| \leq (|P_\alpha||P_\beta| - 1) + |P_\gamma|$. $\square$

Application.

- $s \geq 5$. At least $s - 2 \geq 3$ blocks avoid $\{2, 3\}$. Among three of them at most one pair is split by $\{5, 7\}$, so we choose a pair that is not split. Then $|O| \leq |P_\alpha||P_\beta|$ with $|P_\alpha| + |P_\beta| \leq r - (s - 2)$.
- $s = 4$. If 2 and 3 lie in different blocks, exactly two blocks avoid $\{2, 3\}$. Otherwise at least three do. Either Lemma 9(1) applies with $|P_\alpha| + |P_\beta| \leq r - 2$, or (2) applies with $|P_\alpha| + |P_\beta| + |P_\gamma| \leq r - 1$.

- $s = 3$, where 2 and 3 do not lie in two different blocks (they share a block, or one of them lies in no block). Then two blocks avoid $\{2, 3\}$, and $|P_\alpha| + |P_\beta| + |P_\gamma| \leq r$.

The script maximises ab subject to $a + b \leq r - (s - 2)$, and $ab - 1 + c$ (when $p \geq 35$), over positive integers. It confirms that the maximum is below $k - s$:

	$s = 4$	5	6	7	8	9	10	11	12	$s = 3$ (Case I)
$k = 32$: bound / $k - s$	16/28	12/27	9/26	6/25	4/24	2/23	1/22	–	–	20/29
$k = 38$: bound / $k - s$	20/34	16/33	12/32	9/31	6/30	4/29	2/28	1/27	–	25/35
$k = 42$: bound / $k - s$	25/38	20/37	16/36	12/35	9/34	6/33	4/32	2/31	1/30	30/39

For instance, for $k = 42$ and $s = 4$ we get $\max(5 \cdot 5, 5 \cdot 5 - 1 + 1) = 25 < 38$; and for $s = 3$ we get $\max_{a+b+c \leq 12}(ab + c - 1) = 5 \cdot 6 + 0 = 30 < 39$.

Case II: $s = 3$ with 2 and 3 in different blocks

Renumber so that $2 \in P_1$ and $3 \in P_2$. Then P_3 contains neither 2 nor 3. Write $a = |P_1|$, $b = |P_2|$, $c = |P_3|$, so $a + b + c \leq r$.

The choice function. Put $R = \{2, 3\} \cup (\{5\} \cap P_3)$. (If 5 lies in P_1 or P_2 , it is *not* in R .) Define $\omega_j(l) = \min(\pi(l) \cap P_j \setminus R)$ if this set is nonempty, and $\omega_j(l) = \min(\pi(l) \cap P_j)$ otherwise. Since $R \cap P_1 = \{2\}$, $R \cap P_2 = \{3\}$ and $R \cap P_3 \subseteq \{5\}$, we get:

$$\omega_1(l) = 2 \iff \pi(l) \cap P_1 = \{2\}, \quad \omega_2(l) = 3 \iff \pi(l) \cap P_2 = \{3\}, \quad \omega_3(l) = 5 \iff \pi(l) \cap P_3 = \{5\} \quad (2)$$

(the last one when $5 \in P_3$).

Lemma 10 (multiplicity of $(2, 3, 5)$). *Suppose $5 \in P_3$, and let $L = \{l \in O : \omega(l) = (2, 3, 5)\}$. Then $|L| \leq 11$.*

Proof. Let $l \in L$. By (2) and (1), $\text{pr}(g_{l,i_1}) = \{2\}$, $\text{pr}(g_{l,i_2}) = \{3\}$ and $\text{pr}(g_{l,i_3}) = \{5\}$. So $g_{l,i_1} = 2^e$, $g_{l,i_2} = 3^f$ and $g_{l,i_3} = 5^h$, with $e, f, h \geq 1$.

Let $l \neq l'$ in L . By Lemma 8, $g_{ll'} = 30$. By Lemma 4, $a_l \not\equiv a_{l'} \pmod{30}$. Moreover 4 does not divide both m_l and $m_{l'}$, since otherwise $4 \mid g_{ll'}$. The same holds for 9 and for 25. Hence the set

$$E = \{l \in L : 4 \mid m_l \text{ or } 9 \mid m_l \text{ or } 25 \mid m_l\}$$

has at most 3 elements.

For $l \in L \setminus E$, $g_{l,i_1} = 2^e$ divides m_l and $4 \nmid m_l$, so $g_{l,i_1} = 2$. Likewise $g_{l,i_2} = 3$ and $g_{l,i_3} = 5$. Lemma 4 gives $a_l \not\equiv a_{i_1} \pmod{2}$, $a_l \not\equiv a_{i_2} \pmod{3}$ and $a_l \not\equiv a_{i_3} \pmod{5}$. By the Chinese remainder theorem, exactly $1 \cdot 2 \cdot 4 = 8$ residues modulo 30 meet these three conditions. The elements of $L \setminus E$ have pairwise distinct residues modulo 30, so $|L \setminus E| \leq 8$ and $|L| \leq 8 + 3 = 11$. $\square$

Put $\varepsilon = 10$ if $5 \in P_3$ and $\varepsilon = 0$ otherwise, so that $\varepsilon \geq |L| - 1$ whenever L is defined. Let $\Lambda = \{(y, x) \in P_2 \times P_3 : yx \leq p\}$ and $\Lambda' = \{(q, x) \in (P_1 \setminus \{2\}) \times P_3 : qx \leq p\}$. Call a prime q *small* if $2q \leq p$ and *big* otherwise. Let σ_2 and σ_3 be the numbers of small primes in P_2 and P_3 , and let $\beta_2 = b - \sigma_2$ and $\beta_3 = c - \sigma_3$.

Lemma 11 (first bound). $|O| \leq B_1 := bc + |\Lambda|(a - 1) + \varepsilon$.

Proof. Partition O into the fibres $F(y, x) = \{l : \omega_2(l) = y, \omega_3(l) = x\}$ for $(y, x) \in P_2 \times P_3$. If $(y, x) \notin \Lambda$, then $|F(y, x)| \leq 1$ by Lemma 8.

Let $(y, x) \in \Lambda$. Two elements of $F(y, x)$ with the same ω_1 agree in three coordinates. By Lemma 8 this forces $\{\omega_1, y, x\} = \{2, 3, 5\}$, that is, $y = 3, x = 5$ and $\omega_1 = 2$. So the elements of $F(y, x)$ have pairwise distinct ω_1 , except that the value 2 may repeat in $F(3, 5)$, where the elements with $\omega_1 = 2$ form exactly L . Hence $|F(y, x)| \leq a$, and $|F(3, 5)| \leq (a - 1) + |L| \leq a + \varepsilon$. Summing over all fibres gives $|O| \leq (bc - |\Lambda|) + a|\Lambda| + \varepsilon$. $\square$

Lemma 12 (second bound). $|O| \leq B_2 := \min(bc, \sigma_2\sigma_3 + \beta_2 + \beta_3) + \varepsilon + (a - 1)c + |\Lambda'|(b - 1)$.

Proof. Split $O = T \cup U$, where $T = \{l : \omega_1(l) = 2\}$ and $U = O \setminus T$.

T . Map $l \in T$ to the cell $(\omega_2(l), \omega_3(l)) \in P_2 \times P_3$. Two elements of T in the same cell (y, x) agree in all three coordinates. By Lemma 8 this can happen only for the cell $(3, 5)$, whose content is L . So $|T| \leq bc - 1 + |L| \leq bc + \varepsilon$.

Next, suppose $y \in P_2$ is big. Two elements of T in row y agree on $\{2, y\}$, which is not light (Lemma 7(3)). So row y contains at most one element of T . Likewise, a big column $x \in P_3$ contains at most one element of T . Every element of T lies in a big row, or in a small row and a big column, or in a small row and a small column. Hence $|T| \leq \beta_2 + \beta_3 + \sigma_2\sigma_3 + \varepsilon$, since a cell outside $(3, 5)$ holds at most one element.

U . Map $l \in U$ to $(\omega_1(l), \omega_3(l)) \in (P_1 \setminus \{2\}) \times P_3$. Both coordinates are primes ≥ 5 , because $3 \in P_2$. If $(q, x) \notin \Lambda'$, the fibre has at most one element (Lemma 8). If $(q, x) \in \Lambda'$, two elements of the fibre cannot also agree in ω_2 , since $\{q, x, y\}$ with $q, x \geq 5$ is not light. So the fibre injects into P_2 and has at most b elements. Hence $|U| \leq (a - 1)c + |\Lambda'|(b - 1)$. $\square$

Conclusion of Case II. $|O| \leq \min(B_1, B_2)$. Both bounds depend only on which primes lie in $P_1 \ni 2, P_2 \ni 3$ and P_3 . The script `check_p4_c6.py` enumerates every placement of each prime $5 \leq q < p$ into P_1, P_2, P_3 or no block: 4^{r-2} placements, of which 58 975, 242 461 and 989 527 have $P_3 \neq \emptyset$ for $k = 32, 38, 42$. It evaluates $\min(B_1, B_2)$ in exact integer arithmetic for each one. The maxima are:

k	$\max \min(B_1, B_2)$	$k - 3$	a worst placement
32	26	29	$P_1 = \{2, 17, 19\}, P_2 = \{3, 11, 13\}, P_3 = \{5, 7, 23, 29\}$
38	31	35	$P_1 = \{2, 13, 19, 23, 29, 31\}, P_2 = \{3, 7\}, P_3 = \{5, 11, 17\}$
42	38	39	$P_1 = \{2, 17, 23, 29, 31, 37\}, P_2 = \{3, 7\}, P_3 = \{5, 11, 13, 19\}$

So $|O| < k - 3$ in every placement. We check the worst case for $k = 42$ by hand: $a = 6, b = 2, c = 4$, and $\Lambda = \{(3, 5), (3, 11), (3, 13), (7, 5)\}$. This gives $B_1 = 8 + 4 \cdot 5 + 10 = 38$. Also $\sigma_2 = 2, \sigma_3 = 4$ and $\Lambda' = \emptyset$, so $B_2 = \min(8, 8) + 10 + 5 \cdot 4 + 0 = 38$.

All cases lead to $|O| < k - s = |O|$, a contradiction. This proves Theorem 1. $\square$

4 The size $k = 44$ (computer-assisted, certificate incomplete)

Now $p = 43$ and $r = 13$. We follow Sections 2–3 and list only the changes. Suppose the conjecture holds at 43, and the family is disjoint with all $g_{ij} \leq 43$. Define $S, s \geq 3, O$, the blocks and choice functions exactly as before. Lemmas 4–6 are unchanged.

Light sets for $p = 43$. Lemma 7 holds with one change: the light 3-sets are $\{2, 3, 5\}$ and $\{2, 3, 7\}$ ($42 \leq 43$). The other claims are unchanged:

- $210 > 43$ and $2 \cdot 3 \cdot 11 = 66 > 43$, and every other 3-set has product $\geq 2 \cdot 5 \cdot 7 = 70$;

- $5 \cdot 11 = 55 > 43$, so $\{5, 7\}$ is the only light pair of primes ≥ 5 ;
- a light set containing q with $2q > 43$ is $\{q\}$.

The script asserts these facts. Accordingly, Lemma 8 now says: agreement in three coordinates forces $\{2, 3, 5\}$ with $g_{ll'} = 30$, or $\{2, 3, 7\}$ with $g_{ll'} = 42$. Here $30 \mid g$ or $42 \mid g$, and $g \leq 43 < 60$.

Case I ($s \geq 4$; and $s = 3$ with 2, 3 not in two different blocks). Lemma 9 and its application go through verbatim: $\{5, 7, z\}$ is still never light. The resulting bounds (from `check_p4_c6.py`) are, for $s = 4, \dots, 13$: 30, 25, 20, 16, 12, 9, 6, 4, 2, 1, against $k - s = 40, \dots, 31$. For $s = 3$ in Case I the bound is $36 < 41$.

Case II ($s = 3$, $2 \in P_1$, $3 \in P_2$). Let $T_3 = \{5, 7\} \cap P_3$ and $R = \{2, 3\} \cup T_3$, and define ω by the rule of Section 3 with this R . Then $R \cap P_1 = \{2\}$ and $R \cap P_2 = \{3\}$, so the first two equivalences in (2) still hold. For $t \in T_3$ put $L_t = \{l \in O : \omega(l) = (2, 3, t)\}$. For $l \in L_t$ we have $\pi(l) \cap P_3 \subseteq T_3$ and $\min(\pi(l) \cap P_3) = t$. Hence

$$\pi(l) \cap P_3 = \{7\} \text{ for } l \in L_7, \quad \pi(l) \cap P_3 \in \{\{5\}, \{5, 7\}\} \text{ for } l \in L_5.$$

Lemma 13 (multiplicities for $p = 43$). 1. If $5 \in P_3$, then $|L_5| \leq 12$; if moreover $7 \notin P_3$, then $|L_5| \leq 11$.

2. If $7 \in P_3$, then $|L_7| \leq 15$.

3. If $5, 7 \in P_3$, then $|L_5| + |L_7| \leq 18$.

Proof. For $l \in L_5 \cup L_7$, (1) gives $g_{l,i_1} = 2^e$ and $g_{l,i_2} = 3^f$ with $e, f \geq 1$. Also $g_{l,i_3} = 7^h$ if $l \in L_7$, and $g_{l,i_3} = 5^h$ if $l \in L_5$ and $7 \notin \pi(l)$.

(1) Let $l \neq l'$ in L_5 . Then $g_{ll'} = 30$, so $a_l \not\equiv a_{l'} \pmod{30}$. Also $7 \nmid g_{ll'}$, so at most one element of L_5 has $7 \in \pi(l)$ (since $7 \in \pi(l)$ implies $7 \mid m_l$). Likewise at most one element has $4 \mid m_l$, at most one has $9 \mid m_l$, and at most one has $25 \mid m_l$. Call l *exceptional* if $4 \mid m_l$, $9 \mid m_l$ or $25 \mid m_l$, or if $7 \in P_3$ and $7 \in \pi(l)$. There are at most 4 exceptional elements, and at most 3 if $7 \notin P_3$. (If $7 \notin P_3$ then $\pi(l) \cap P_3 = \{5\}$, so 7 plays no role in g_{l,i_3} .) A non-exceptional l has $g_{l,i_1} = 2$, $g_{l,i_2} = 3$ and $g_{l,i_3} = 5$, so $a_l \bmod 30$ lies in a fixed set of $1 \cdot 2 \cdot 4 = 8$ residues. Hence $|L_5| \leq 8 + 4$, and $|L_5| \leq 8 + 3$ if $7 \notin P_3$.

(2) Let $l \neq l'$ in L_7 . Then $g_{ll'} = 42$. The exceptional elements, those with $4 \mid m_l$, $9 \mid m_l$ or $49 \mid m_l$, number at most 3. A non-exceptional l has $g_{l,i_3} = 7$, so $a_l \bmod 42$ lies in a fixed set of $1 \cdot 2 \cdot 6 = 12$ residues, pairwise distinct. So $|L_7| \leq 15$.

(3) Let $\mathcal{R}$ be the set of the two residues $x \bmod 6$ with $x \not\equiv a_{i_1} \pmod{2}$ and $x \not\equiv a_{i_2} \pmod{3}$. Every non-exceptional element of $L_5 \cup L_7$ has $a_l \bmod 6 \in \mathcal{R}$.

Claim. If $l \in L_5$ with $7 \notin \pi(l)$ and $l' \in L_7$ is non-exceptional, then $g_{ll'} = 6$, and hence $a_l \not\equiv a_{l'} \pmod{6}$.

Proof of the claim. By Lemma 5, $\text{pr}(g_{ll'}) = \pi(l) \cap \pi(l')$, which contains 2 and 3. We have $5 \notin \pi(l')$, because $\pi(l') \cap P_3 = \{7\}$ and $5 \in P_3$. Also $7 \notin \pi(l)$. Any other shared prime $q \geq 11$ would give $6q > 43$. So $\text{pr}(g_{ll'}) = \{2, 3\}$. Since $4 \nmid m_{l'}$ and $9 \nmid m_{l'}$, we get $g_{ll'} = 6$, and Lemma 4 applies.

Now let $n_5(\rho)$ and $n_7(\rho)$ be the numbers of non-exceptional elements of L_5 and L_7 with residue $\rho \in \mathcal{R}$ modulo 6. By the claim, $n_5(\rho)n_7(\rho) = 0$. By (1) and (2), $n_5(\rho) \leq 4$ (distinct residues mod 30 above ρ , avoiding $a_{i_3} \bmod 5$) and $n_7(\rho) \leq 6$.

- *Case A:* $n_7(\rho) > 0$ for both $\rho \in \mathcal{R}$. Then no non-exceptional element of L_5 exists. Let $l \in L_5$ with $7 \notin \pi(l)$. By the claim, $a_l \bmod 6$ differs from both elements of $\mathcal{R}$. So $a_l \equiv a_{i_1} \pmod{2}$, which needs $g_{l,i_1} \neq 2$ and hence $4 \mid m_l$; or $a_l \equiv a_{i_2} \pmod{3}$, which needs $9 \mid m_l$. At most one element of L_5 has $4 \mid m_l$, at most one has $9 \mid m_l$, and at most one has $7 \in \pi(l)$. So $|L_5| \leq 3$ and $|L_5| + |L_7| \leq 3 + 15 = 18$.
- *Case B:* $n_7(\rho_0) = 0$ for some $\rho_0 \in \mathcal{R}$. The non-exceptional elements number at most $n_5(\rho_0) + \max(n_5(\rho_1), n_7(\rho_1)) \leq 4 + 6 = 10$. Adding at most $4 + 3$ exceptional elements gives $|L_5| + |L_7| \leq 17$. $\square$

Put $\varepsilon = 10, 14, 16$ according as $T_3 = \{5\}, \{7\}, \{5, 7\}$, and $\varepsilon = 0$ if $T_3 = \emptyset$. By Lemma 13, $\sum_{t \in T_3} \max(|L_t| - 1, 0) \leq \varepsilon$ in each case. For $\{5, 7\}$ this follows from $|L_5| + |L_7| \leq 18$, $|L_5| \leq 12$ and $|L_7| \leq 15$.

Lemmas 11 and 12 hold verbatim with this ε . In their proofs the only fibres, or cells, holding a repeated ω -value are $(3, t)$ with $t \in T_3$, whose repeated part is L_t . Also, $\{q, x, y\}$ with $q, x \geq 5$ is still never light.

The code relaxation. Consider the multiset $\Phi = \{\omega(l) : l \in O\}$ of tuples in $P_1 \times P_2 \times P_3$. It satisfies the following.

- (C1) If $l \neq l'$ and $\omega(l) \neq \omega(l')$, the primes on which they agree form a light set (Lemma 6).
- (C2) Only the tuples $(2, 3, t)$ with $t \in T_3$ can occur more than once (Lemma 8).
- (C3) Their multiplicities obey Lemma 13.

So $|O| = |\Phi|$ is at most the largest size N of a multiset obeying (C1)–(C3).

The computation (`sat_residual.py`, output in `sat_residual.log`).

1. Enumerate the placements of the primes $5, \dots, 41$ into P_1, P_2, P_3 or no block, with $P_3 \neq \emptyset$. The primes $17, 19$ satisfy $2q \leq 43 < 3q$; call them the *H*-primes. The primes $23, \dots, 41$ satisfy $2q > 43$; call them the *B*-primes.

Permuting the H-primes among themselves, or the B-primes among themselves, preserves lightness of every set. A set containing an H-prime h is light iff it is $\{h\}$ or $\{2, h\}$; one containing a B-prime is light iff it is a singleton. So such a permutation preserves B_1, B_2 and N . It therefore suffices to place $5, 7, 11, 13$ individually and only the *numbers* of H- and B-primes in each block: 133 154 placements.

2. In 132 673 of them, $\min(B_1, B_2) \leq 40 < 41 = k - 3$.
3. For each of the remaining 481, a SAT solver (CaDiCaL via PySAT) proves that no multiset obeying (C1)–(C3) has size ≥ 41 . Each tuple gets a Boolean variable, and each incompatible pair gets a binary clause. A repeated tuple $(2, 3, t)$ gets 15 or 12 “copy” variables that imply it. A cardinality constraint caps the copies of $(2, 3, 5)$ and $(2, 3, 7)$ together at 18, and another requires at least 41 counted literals. All 481 instances are UNSAT.

Hence $|O| \leq 40 < 41$ in Case II, which is the reduction behind Remark 2. $\square$

How the joint bound was needed. With the weaker joint bound 19 (from (1)+(2) with only the Case B argument), exactly 8 placements are not excluded by the SAT relaxation, for example $P_1 = \{2, 11, 13\}$, $P_2 = \{3, 17, 19\}$, $P_3 = \{5, 7, 23, 29, 31, 37, 41\}$. There the relaxation reaches exactly $41 = k - 3$. With 18, the relaxation stays ≤ 40 on all of them.

5 Remarks

1. *Where the hypothesis enters.* The hypothesis is used only in Step 2 ($s \geq 3$), and only at size $k - 1$.
2. *Why Cell 5's argument breaks.* For $p \geq 31$ the full map ω need not be injective, because two classes may share exactly the primes $\{2, 3, 5\}$ ($\gcd = 30$). AM–GM alone also fails, independently of injectivity. For $k = 32$ it allows $(3, 3, 2, 2)$ with product $36 \geq 28$ at $s = 4$, and $(2, 2, 2, 2, 2)$ with $32 \geq 27$ at $s = 5$. Lemma 9 replaces the s -fold product by a product of two coordinates.
3. *Tightness.* For $k = 42$ the margin is one ($38 < 39$), and it uses the full strength of Lemma 10.
4. $k = 48$ ($p = 47$) *is not decided.* All of Case I and $s \geq 4$ closes (bounds $36, 30, \dots, 1$ against $44, 43, \dots, 34$, and $42 < 45$ for $s = 3$ in Case I). The multiplicity lemma, Lemma 13, holds verbatim for $p = 47$: every step uses only $6q > p$ for $q \geq 11$ and $g < 60$. Case II is **not** closed, however: in a partial run (`sat_residual_48_partial.log`, stopped for time) the code relaxation (C1)–(C3) already reaches $45 = k - 3$ on at least two placements, for example $P_1 = \{2, 11, 41, 43\}$, $P_2 = \{3, 17, 19, 23\}$, $P_3 = \{5, 7, 13, 29, 31, 37\}$. Closing $k = 48$ needs information beyond (C1)–(C3), such as residue interactions between L_5 , L_7 and the other cells $(2, 3, x)$ or $(2, y, 5)$, which have $\gcd 6$ or 10 with them. The sizes $k = 54, 60, \dots$ were not attempted.

Files (directory `p4_c6/`).

- `check_p4_c6.py`: every inequality of Sections 3 and 4 (Case I); output in `check_p4_c6.log`. This enumeration is part of the proof for $k = 32, 38, 42$ (the maxima in Case II are computed over all placements, not derived by hand), so these three sizes are *computer-assisted*. The set enumerated is the finite set of prime distributions defined in Section 3; it runs in 4 s on a laptop, and an independent rerun (`check_p4_c6_rerun.log`) reproduces the output exactly.
- `sat_residual.py`: Case II for $k = 44$; output in `sat_residual.log`.
- `validate.py`, `validate44.py`: self-review cross-checks. On random placements the exact code optimum never exceeds $\min(B_1, B_2)$. The SAT encoding is not vacuous: it is satisfiable at 40 and unsatisfiable at 41 on a residual placement.
- `probe_joint.py`: the joint bound 19 versus 18 for $k = 44$.